

Comandos de arranque y rescate ARINC

Referencia corta para notebook, Prometheus/Grafana, Raspberry Pi 3B+, bridge SPI y dashboard Flask.

Idea clave

En operacion normal la 3B+ arranca sola con systemd. En la notebook, Prometheus/Grafana se inician con el script PowerShell del repo.

1. Notebook - iniciar Prometheus/Grafana

```
cd "C:\Users\usuario\RASPI PICO\PAMPA\ARINC_SPI\prometheus"
powershell -ExecutionPolicy Bypass -File .\start_prometheus_grafana.ps1
```

2. Notebook - verificar estado

```
cd "C:\Users\usuario\RASPI PICO\PAMPA\ARINC_SPI\prometheus"
powershell -ExecutionPolicy Bypass -File .\start_prometheus_grafana.ps1 -Status
```

3. Notebook - detener Prometheus/Grafana

```
cd "C:\Users\usuario\RASPI PICO\PAMPA\ARINC_SPI\prometheus"
powershell -ExecutionPolicy Bypass -File .\start_prometheus_grafana.ps1 -Stop
```

URLs notebook

Prometheus: <http://127.0.0.1:9090> | Targets: <http://127.0.0.1:9090/targets> | Grafana: <http://127.0.0.1:3000>

4. 3B+ - verificar que el autoarranque funcione

```
systemctl status arinc-sniffer-bridge arinc-dashboard-flask --no-pager
systemctl status arinc-supervisor --no-pager
curl http://127.0.0.1:5100/stats
curl http://127.0.0.1:5100/supervisor/status
curl http://127.0.0.1:5000/stats
```

5. 3B+ - rescate rapido si Flask o bridge no arrancaron

```
sudo systemctl restart arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor
sleep 2
systemctl status arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor --no-pager
```

6. 3B+ - verificar perfil real del bridge

```
PID="$(systemctl show -p MainPID --value arinc-sniffer-bridge)"
tr '\0' '\n' < "/proc/$PID/environ" | grep '^ARINC_SPI_'
```

7. 3B+ - valores esperados del perfil

```
ARINC_SPI_HZ=8000000
ARINC_SPI_TRANSFER_MODE=pio-frame
ARINC_SPI_MANUAL_CS=1
ARINC_SPI_REQUEST_MODE=drdy
ARINC_SPI_DRDY_GPIO=25
```

8. 3B+ - configurar IP fija Ethernet

```
sudo bash ~/PAMPA/HOST_3b+/configurar_eth0_ip_fija.sh
ip -4 addr show eth0
```

9. 3B+ - reinstalar servicios una sola vez

```
cd ~/PAMPA/HOST_3b+

sudo cp systemd/arinc-sniffer-bridge.service /etc/systemd/system/
sudo cp systemd/arinc-dashboard-flask.service /etc/systemd/system/
sudo cp systemd/arinc-supervisor.service /etc/systemd/system/

sudo systemctl daemon-reload
```

10. 3B+ - habilitar autoarranque

```
sudo systemctl enable arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor
sudo systemctl start arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor
```

11. 3B+ - detenerlos ahora

```
sudo systemctl stop arinc-supervisor arinc-dashboard-flask arinc-sniffer-bridge
```

12. 3B+ - evitar autoarranque en el futuro

```
sudo systemctl disable arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor
```

13. 3B+ - volver a habilitar autoarranque

```
sudo systemctl enable arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor
```

14. 3B+ - ver logs en vivo

```
journalctl -u arinc-sniffer-bridge -f
journalctl -u arinc-dashboard-flask -f
journalctl -u arinc-supervisor -f
```

15. 3B+ - matar procesos manuales si quedaron duplicados

```
pkill -f '[s]pi_sniffer_bridge.py'
pkill -f '[a]rinc_dashboard/./app.py'
sudo systemctl restart arinc-sniffer-bridge arinc-dashboard-flask arinc-supervisor
```

Acceso operativo

Con Ethernet directa: notebook 192.168.50.1/24, 3B+ 192.168.50.2/24. Dashboard: <http://192.168.50.2:5000>.